Sound Calibration: Quick Results

Rice Auditory Benchmarking | 28 October 2026

Setup. An Audient iD14 played 10 ms tone pips; an Earthworks M50 microphone recorded them approximately 10 cm from the speaker. A REED R8090 at 114 dB provided the reference.

Correction. We lowered the playback level of stronger frequencies so the recorded sound components were closer in level. In two tests, the range across the 14 frequency-group medians was **1.13 dB** and **0.77 dB**.

Results from the second test

Frequency (kHz)	Estimated level (dB)	Frequency (kHz)	Estimated level (dB)
2	41.3	18.2	41.6
4	42.1	19.5	41.8
6	41.6	20.9	41.6
8	41.6	22.4	41.7
9	41.7	24	41.7
10.5	41.9	25.8	41.7
17	41.6	27	41.9

Levels are group medians for the frequency component in the middle 6 ms of each pip, expressed as equivalent-sine levels relative to 20 μPa. Estimates assume the nominal 114 dB reference, unchanged linear gain, and a flat microphone response.

Calibration details and measurements